Parametric Modelling of Diffusion MRI with Uncertainty Estimation (Figures)

Computational Modelling for Biomedical Imaging

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1 Parameter Estimation and Mapping

1.1 Ball-and-Stick Model Fitting

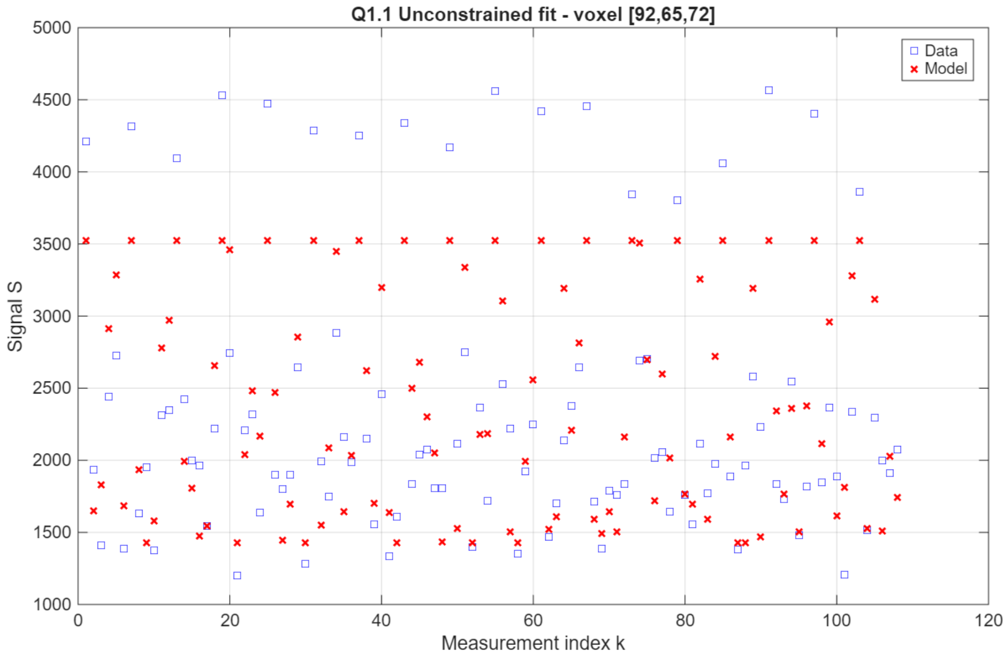


Figure 1: Ball-and-stick model fit versus measured signals for voxel $[92, 65, 72]$, unconstrained optimisation.

Table 1: Estimated parameters of the unconstrained ball-and-stick model fit.

S_0	d	f	θ (rad)	ϕ (rad)	RESNORM
3.5239×10^3	-4.8868×10^{-6}	-1.2256×10^2	8.8104×10^{-1}	1.5531	3.0640×10^7

1.2 Parameter Constraints via Transformation

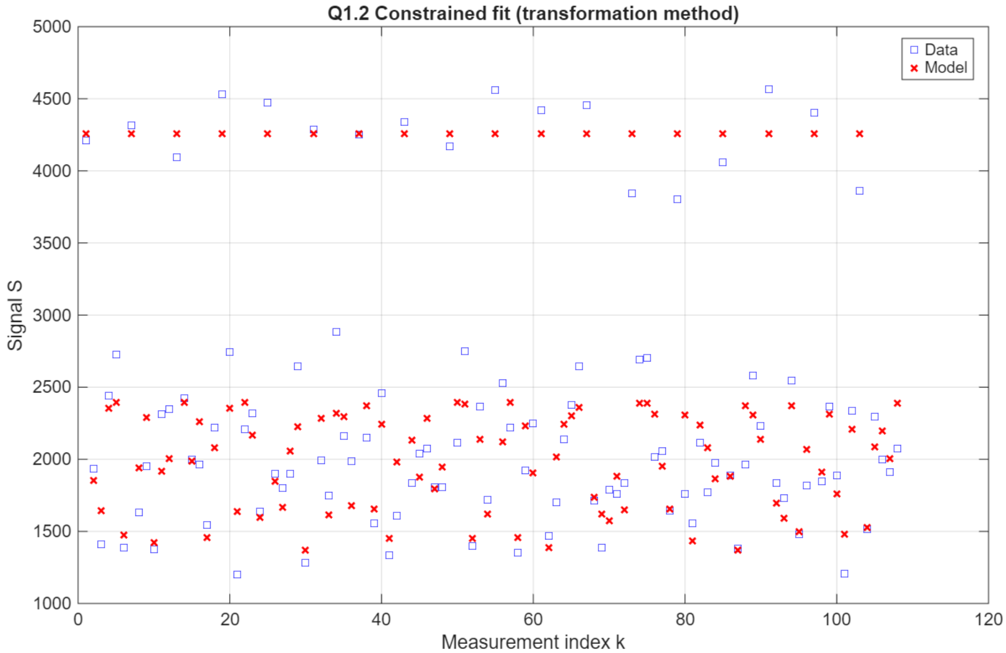


Figure 2: Constrained ball-and-stick model fit versus measured signals for voxel [92, 65, 72].

Table 2: Estimated parameters of the constrained ball-and-stick model fit.

S_0	d	f	θ (rad)	ϕ (rad)	RESNORM
4257.92	1.1413×10^{-3}	0.3573	-0.9811	0.5794	5.8720×10^{6}

1.3 Global Minimum Analysis

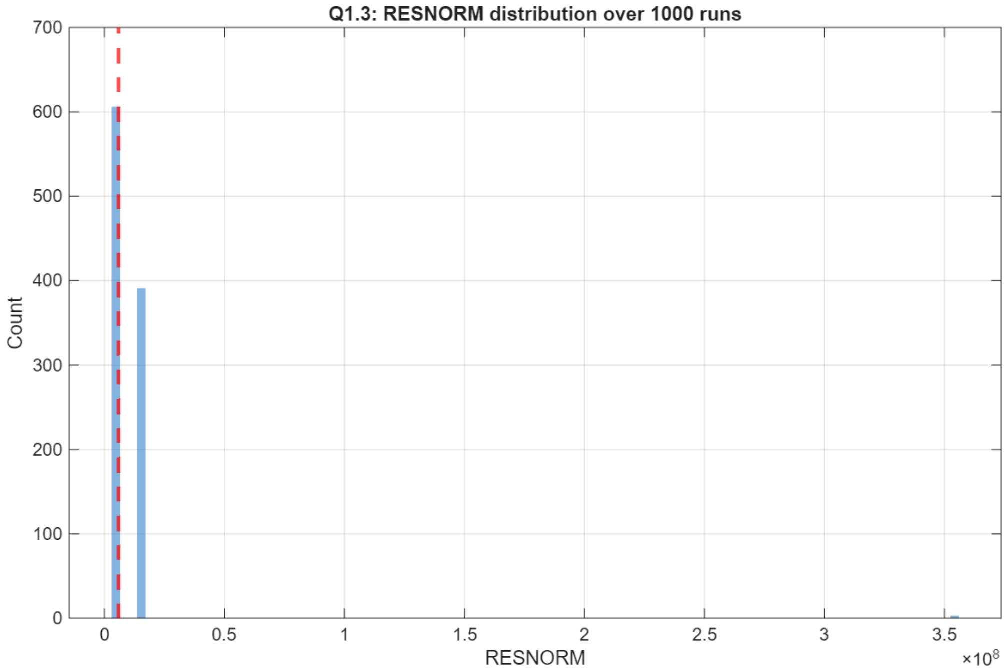


Figure 3: Histogram of RESNORM values across 1000 multi-start optimisation runs for voxel $[92, 65, 72]$, with the candidate global minimum indicated.

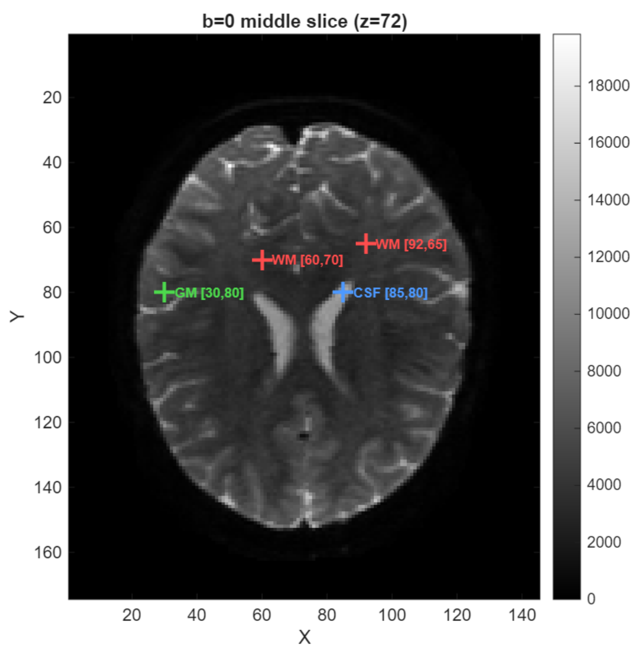


Figure 4: b = 0 diffusion MRI image of the middle slice (z = 72) with voxels used in the analysis highlighted.

Table 3: Estimated parameters and optimisation statistics for the analysed voxels.

Voxel	RESNORM	p_{global}	N_{95}	S_0	d (mm^2/s)	f	θ	ϕ
WM [92,65]	5.872×10^6	0.61	4	4257.9	1.1413×10^{-3}	0.3573	-0.9811	0.5794
WM [60,70]	4.835×10^6	0.67	3	3704.8	1.4525×10^{-3}	0.5836	-0.8978	0.4170
GM [30,80]	4.072×10^6	0.83	2	4626.3	9.8291×10^{-4}	0.2048	-0.8675	2.0650
CSF [85,80]	4.375×10^6	0.20	14	11897.3	2.9539×10^{-3}	0.0087	1.1084	-2.1670

1.4 Parameter Mapping

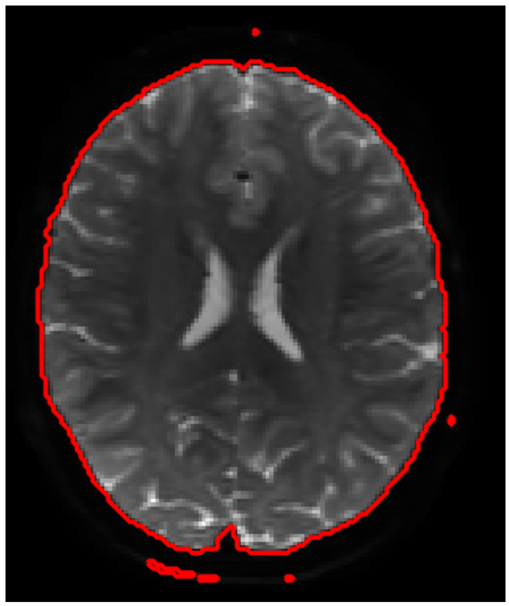


Figure 5: Brain mask obtained from the mean $b = 0$ image using intensity thresholding. Voxels inside the mask (red contour) were included in the parameter estimation.

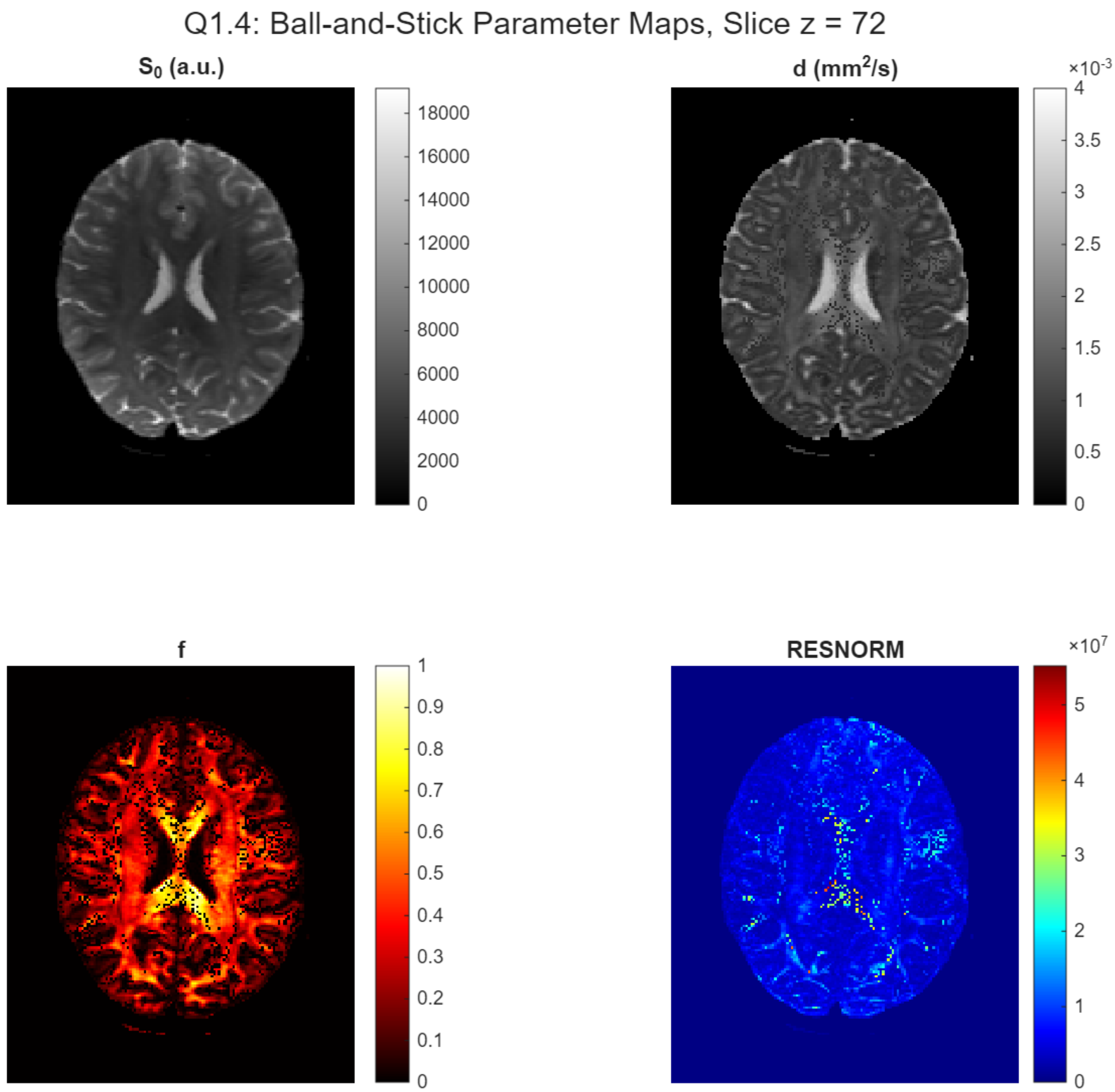
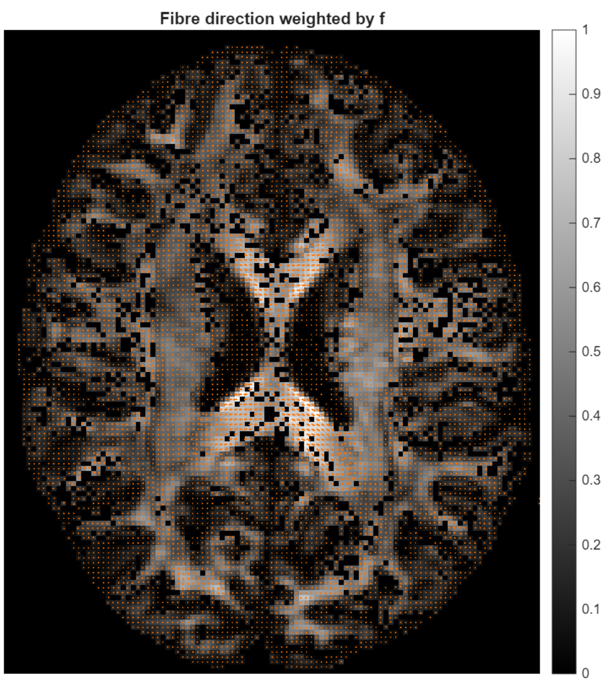
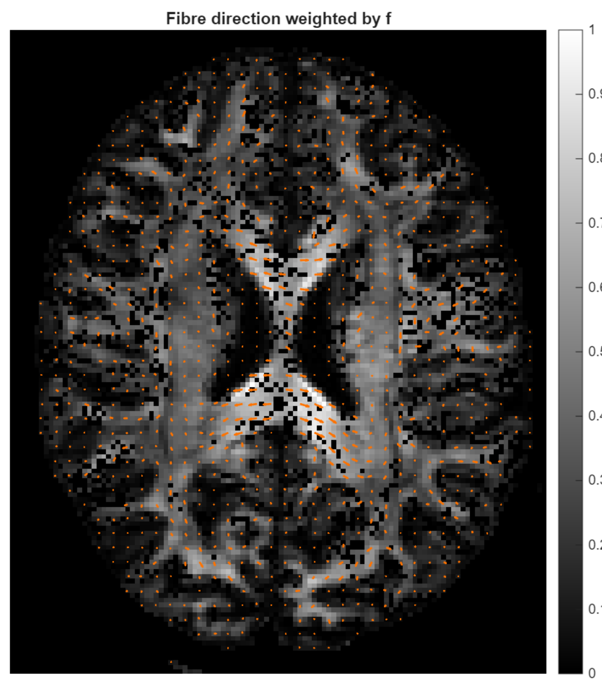


Figure 6: Estimated ball-and-stick parameter maps for the middle axial slice (z = 72): baseline signal S_0, diffusivity d, stick fraction f, and residual sum of squares (RESNORM).



(a) Quiver plot of the fibre direction field $f\mathbf{n}$ with arrows shown at every voxel (step size $= 1$).



(b) Subsampled quiver plot (step size $= 3$), showing one arrow per 3×3 voxel block..

Figure 7: Estimated fibre orientations weighted by the stick fraction f

1.5 Improving Optimisation

Table 4: Comparison of different DT-informed initialisation strategies for the ball-and-stick model.

Mapping	p_{global}	N_{95}	t_{DT} (s)	t_{run} (s)	Slice (min)
M1: λ_1 + FA + $\mathbf{e}_1$	0.781	2	0.01843	0.0029	4.2
M2: MD + FA + $\mathbf{e}_1$	0.793	2	0.01843	0.0029	4.2
M3: λ_1 + 0.5 + $\mathbf{e}_1$	0.771	3	0.01843	0.0029	4.7
M4: λ_1 + FA + zeros	0.699	4	0.01843	0.0031	5.3
M5: MD + 0.5 + zeros	0.619	4	0.01843	0.0032	5.4

Table 5: Comparison of optimisation strategies for the ball-and-stick model.

Method	p_{global}	N_{95}	Time/run (s)	Estimated slice time (min)
Q1.3 Random starts (fminunc + transform)	0.61	4	0.003	1.9
Q1.5.1 Bounds (fmincon)	0.51	5	0.077	66.4
Q1.5.2 Best DT-informed start (MD, FA, $\mathbf{e}_1$)	0.79	2	0.003	4.2

2 Uncertainty Estimation

2.1 Bootstrap Estimation

Table 6: Bootstrap comparison across voxels for parameter S_0.

Voxel	Mean ± 2std	2-sigma range	95% range
WM [92,65]	$4.2562 \times 10^3 \pm 1.1275 \times 10^2$	$[4.1435 \times 10^3, 4.3690 \times 10^3]$	$[4.1370 \times 10^3, 4.3608 \times 10^3]$
WM [60,70]	$3.7040 \times 10^3 \pm 8.0602 \times 10^1$	$[3.6234 \times 10^3, 3.7846 \times 10^3]$	$[3.6288 \times 10^3, 3.7874 \times 10^3]$
GM [30,80]	$4.6271 \times 10^3 \pm 6.9130 \times 10^1$	$[4.5580 \times 10^3, 4.6962 \times 10^3]$	$[4.5589 \times 10^3, 4.6909 \times 10^3]$
CSF [85,80]	$1.1895 \times 10^4 \pm 1.5963 \times 10^2$	$[1.1735 \times 10^4, 1.2054 \times 10^4]$	$[1.1746 \times 10^4, 1.2058 \times 10^4]$

Table 7: Bootstrap comparison across voxels for parameter d.

Voxel	Mean ± 2std	2-sigma range	95% range
WM [92,65]	$1.1394 \times 10^{-3} \pm 1.0604 \times 10^{-4}$	$[1.0334 \times 10^{-3}, 1.2455 \times 10^{-3}]$	$[1.0787 \times 10^{-3}, 1.2045 \times 10^{-3}]$
WM [60,70]	$1.4514 \times 10^{-3} \pm 1.4195 \times 10^{-4}$	$[1.3095 \times 10^{-3}, 1.5934 \times 10^{-3}]$	$[1.3722 \times 10^{-3}, 1.5377 \times 10^{-3}]$
GM [30,80]	$9.8726 \times 10^{-4} \pm 4.2437 \times 10^{-5}$	$[9.4482 \times 10^{-4}, 1.0297 \times 10^{-3}]$	$[9.4453 \times 10^{-4}, 1.0268 \times 10^{-3}]$
CSF [85,80]	$2.9638 \times 10^{-3} \pm 1.1872 \times 10^{-4}$	$[2.8451 \times 10^{-3}, 3.0825 \times 10^{-3}]$	$[2.8490 \times 10^{-3}, 3.0759 \times 10^{-3}]$

Table 8: Bootstrap comparison across voxels for parameter f.

Voxel	Mean ± 2std	2-sigma range	95% range
WM [92,65]	$3.5553 \times 10^{-1} \pm 9.2854 \times 10^{-2}$	$[2.6267 \times 10^{-1}, 4.4838 \times 10^{-1}]$	$[3.1188 \times 10^{-1}, 4.0394 \times 10^{-1}]$
WM [60,70]	$5.8179 \times 10^{-1} \pm 9.9679 \times 10^{-2}$	$[4.8211 \times 10^{-1}, 6.8147 \times 10^{-1}]$	$[5.4151 \times 10^{-1}, 6.2637 \times 10^{-1}]$
GM [30,80]	$2.0819 \times 10^{-1} \pm 4.0261 \times 10^{-2}$	$[1.6793 \times 10^{-1}, 2.4845 \times 10^{-1}]$	$[1.6789 \times 10^{-1}, 2.4558 \times 10^{-1}]$
CSF [85,80]	$9.6600 \times 10^{-3} \pm 1.3006 \times 10^{-2}$	$[-3.3458 \times 10^{-3}, 2.2666 \times 10^{-2}]$	$[3.4298 \times 10^{-17}, 2.0232 \times 10^{-2}]$

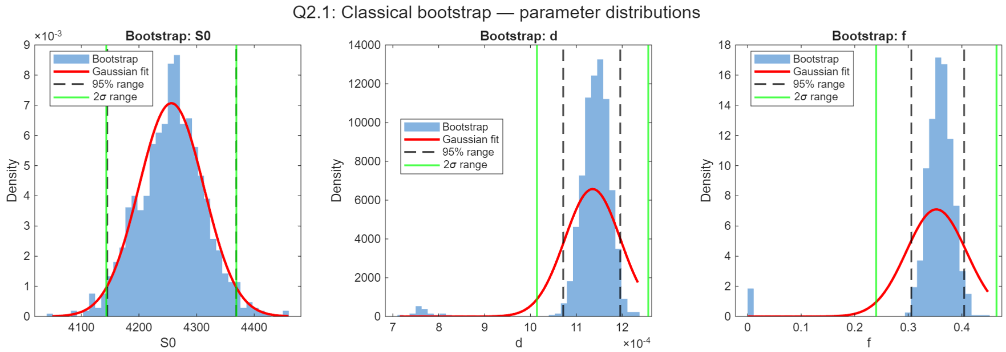


Figure 8: Bootstrap parameter distributions for S_0, d, and f for voxel $[92, 65, 72]$. Histograms show the empirical bootstrap distributions ($T = 1000$ samples). The red curve denotes a Gaussian fit using the bootstrap mean and standard deviation. Black dashed lines indicate the empirical 95% percentile range, while green lines show the 2σ interval ($\mu \pm 2\sigma$).

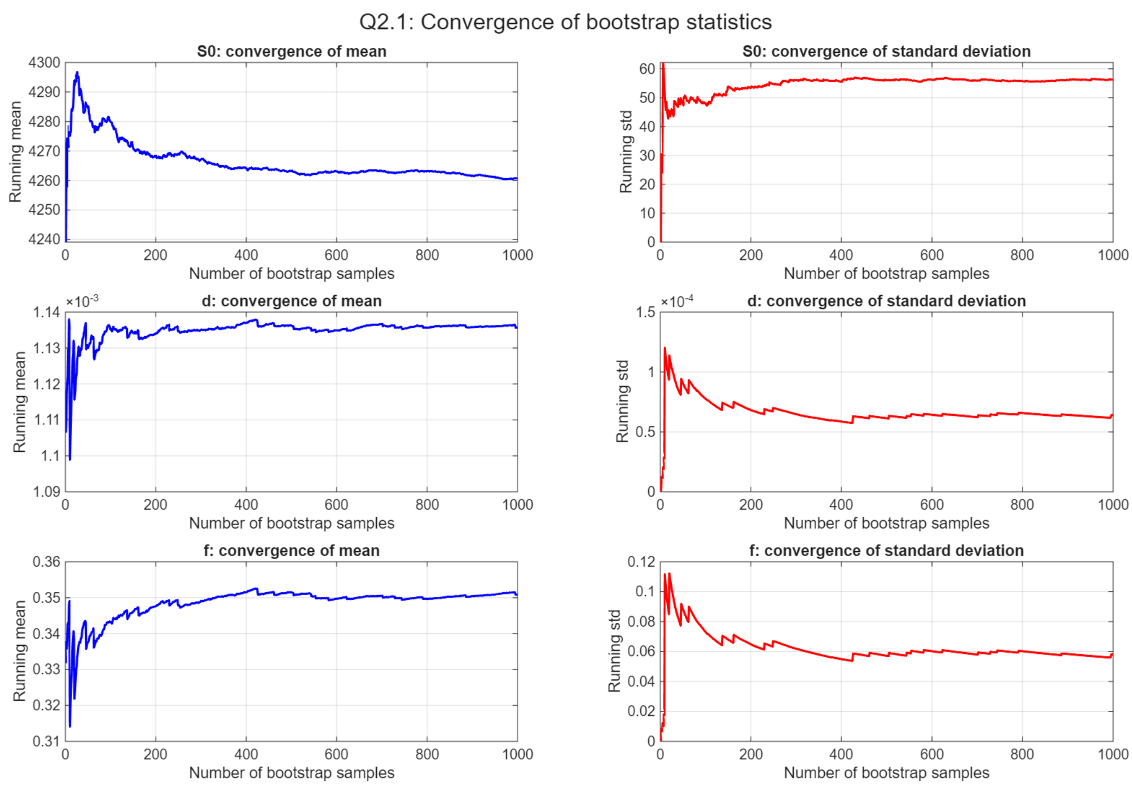


Figure 9: Convergence of bootstrap statistics for parameters S_0, d, and f as a function of the number of bootstrap samples.

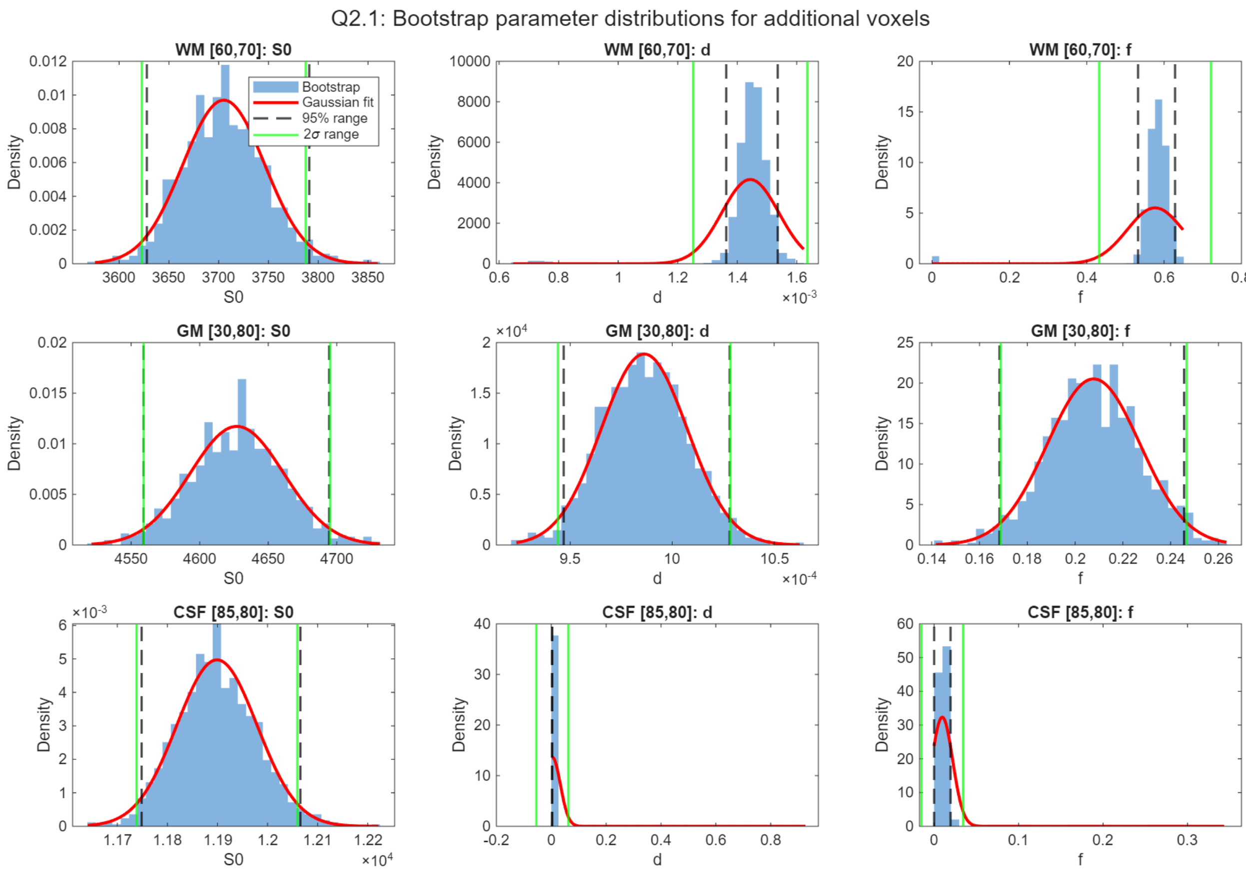


Figure 10: Bootstrap distributions of S_0, d, and f for additional voxels in white matter (WM), grey matter (GM), and cerebrospinal fluid (CSF). Histograms show bootstrap samples with Gaussian fits (red), 95% percentile intervals (black dashed), and 2σ ranges (green).

2.2 MCMC (Metropolis-Hastings)

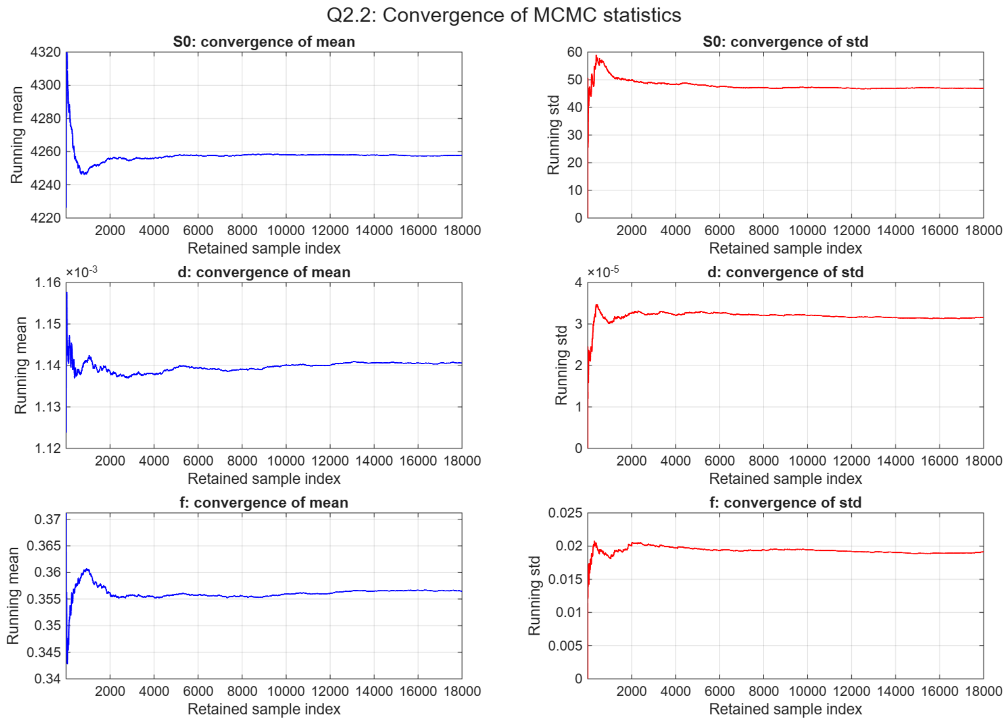


Figure 11: Convergence of MCMC running mean and standard deviation for parameters S_0, d, and f in voxel [92,65,72].

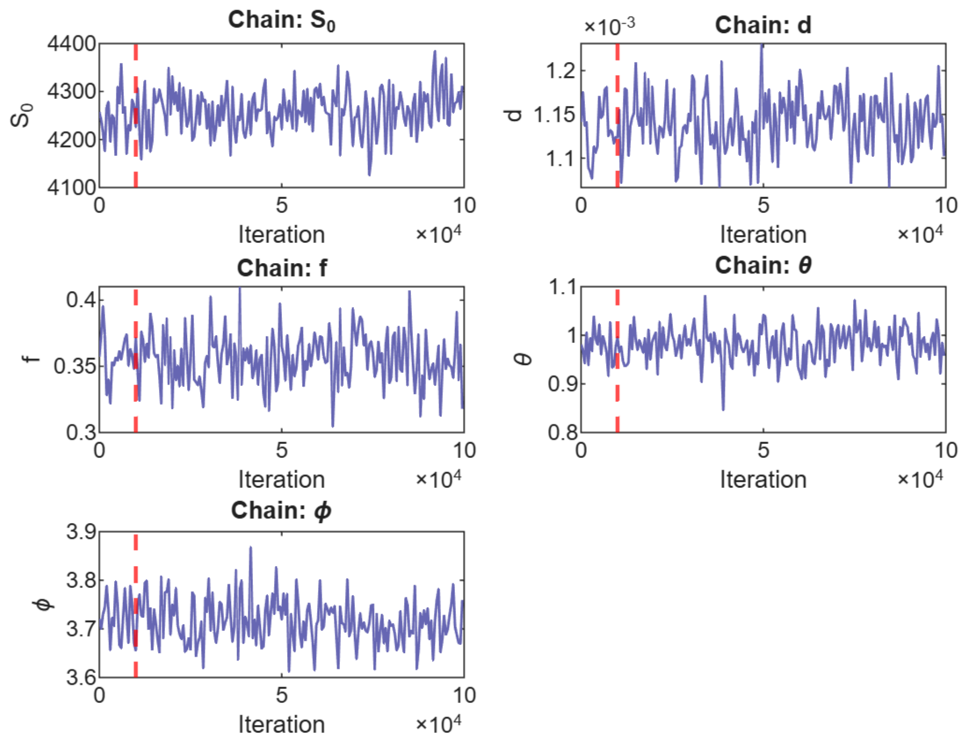


Figure 12: MCMC trace plots for parameters S_0, d, f, θ, and ϕ for voxel [92,65,72]. Every 500th iteration is plotted for clarity. The dashed red line indicates the burn-in period.

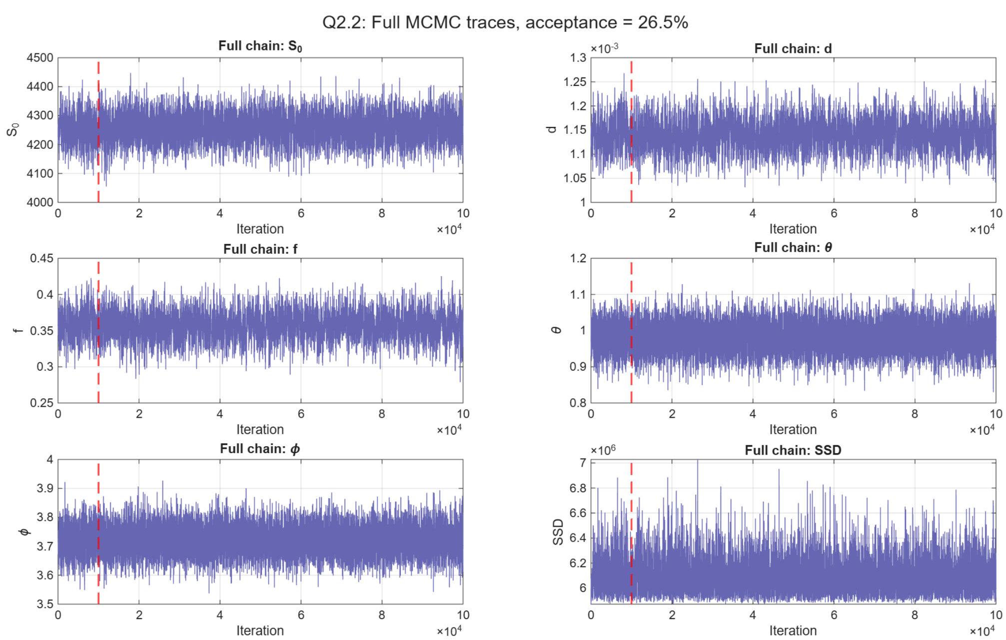


Figure 13: Full MCMC traces for parameters S_0, d, f, θ, and ϕ and SSD for voxel [92,65,72], with the burn-in period indicated by the dashed red line.

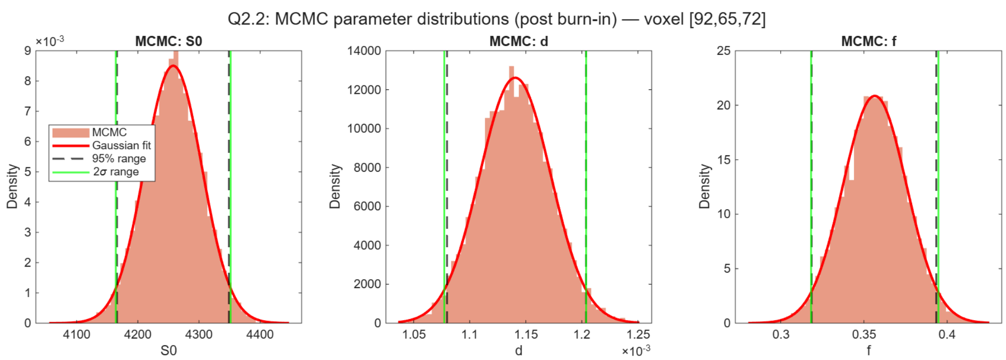


Figure 14: MCMC posterior distributions for parameters S_0, d, and f after burn-in for voxel [92,65,72], with Gaussian fits and uncertainty intervals.

Table 9: Comparison of bootstrap and MCMC uncertainty estimates for parameters S_0, d, and f in voxel [92,65,72].

Param	Bootstrap		MCMC	
	Mean ± 2std	95% range	Mean ± 2std	95% range
S_0	$4.2584 \times 10^{3} \pm 1.102 \times 10^{2}$	$[4.1464 \times 10^{3}, 4.3619 \times 10^{3}]$	$4.2578 \times 10^{3} \pm 9.382 \times 10^{1}$	$[4.1659 \times 10^{3}, 4.3489 \times 10^{3}]$
d	$1.1350 \times 10^{-3} \pm 1.342 \times 10^{-4}$	$[8.3899 \times 10^{-4}, 1.2055 \times 10^{-3}]$	$1.1405 \times 10^{-3} \pm 6.326 \times 10^{-5}$	$[1.0798 \times 10^{-3}, 1.2037 \times 10^{-3}]$
f	$3.5024 \times 10^{-1} \pm 1.242 \times 10^{-1}$	$[9.4282 \times 10^{-12}, 4.0837 \times 10^{-1}]$	$3.5649 \times 10^{-1} \pm 3.825 \times 10^{-2}$	$[3.1859 \times 10^{-1}, 3.9345 \times 10^{-1}]$

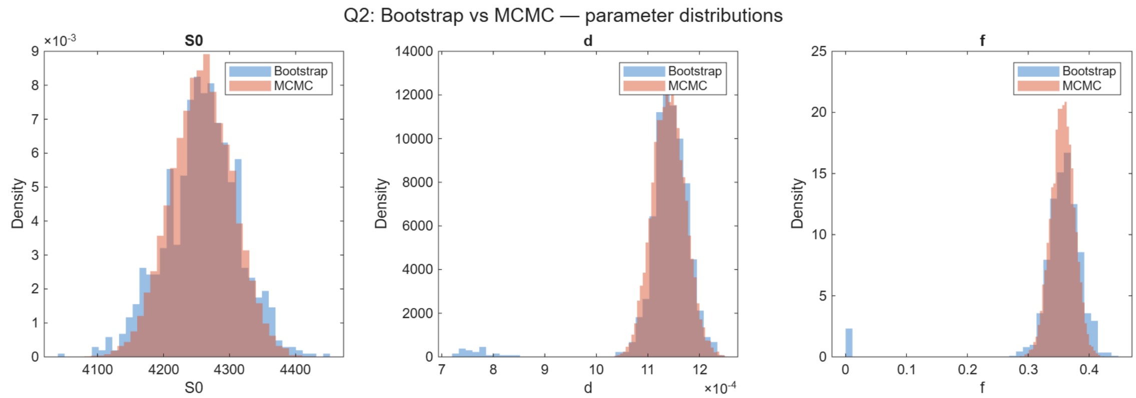


Figure 15: Bootstrap and MCMC parameter distributions for S_0, d, and f in voxel [92,65,72].

3 Model Selection

3.1 Model Fitting to ISBI data

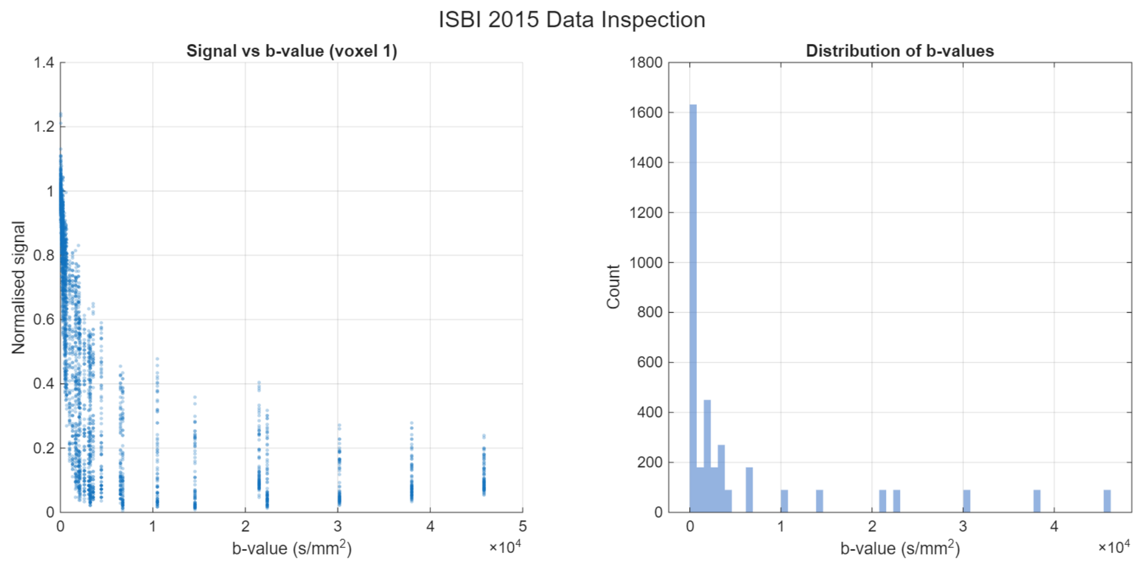


Figure 16: ISBI 2015 data inspection. Normalised signal versus b-value for voxel 1 (left) and distribution of b-values across the dataset (right).

Table 10: DT-informed starting point used for the optimisation.

S_0	d (mm^2/s)	f	θ (rad)	ϕ (rad)
0.7995	0.0001	0.4900	1.5378	3.0505

Table 11: Best-fit parameters from global minimum search (Q3.1).

S_0	d (mm^2/s)	f	θ (rad)	ϕ (rad)	RESNORM	Expected RESNORM
1.0099	1.4321×10^{-3}	0.5749	1.5969	-0.0829	15.106	5.7792

$N_{95} = 1$ — Time per run = 0.012 s

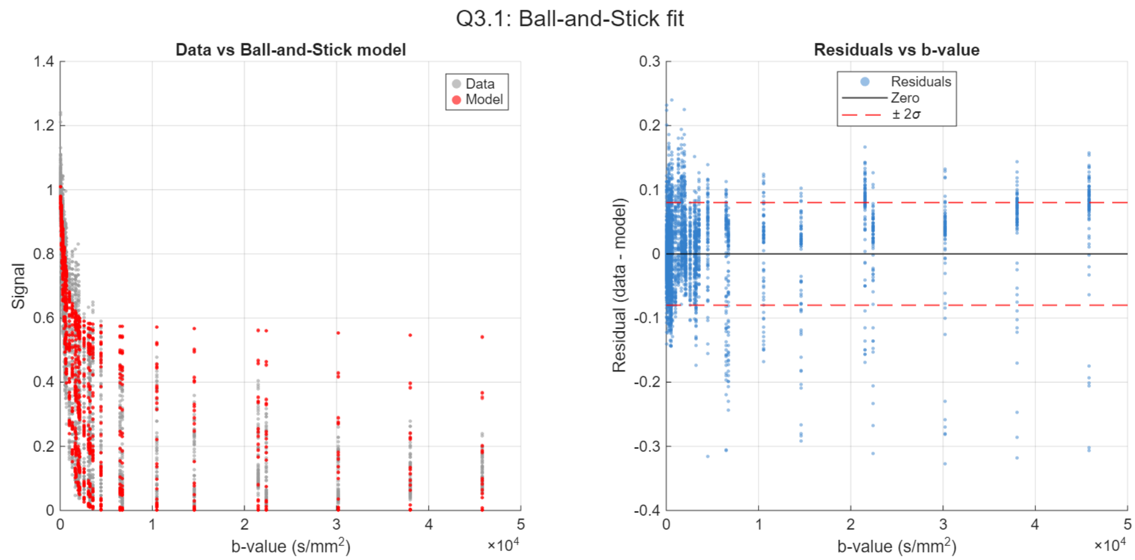


Figure 17: Ball-and-stick model fit. Left: measured diffusion signal and fitted model versus b-value. Right: residuals (data minus model) with $\pm 2\sigma$ bounds indicating the expected noise level.

Figure 18: Normalised diffusion signal as a function of the gradient–fibre angle, where $|\cos(\alpha)| = |\mathbf{q} \cdot \mathbf{n}|$. Colour indicates the b-value, showing stronger attenuation for higher diffusion weightings.

3.2 Other Models

Table 12: Model comparison for Q3.2 based on residual sum of squares.

Model	N_{par}	RESNORM	RN/E[RN]	p_{glob}
Diffusion Tensor	7	223.8511	38.73	linear
Ball-and-Stick	5	15.1060	2.61	1.000
Zeppelin-and-Stick	6	10.8167	1.87	0.999
Zeppelin-Stick-Tortuosity	5	11.6052	2.01	0.934

Expected RESNORM: $K\sigma^2 = 5.7792 \quad (K = 3612,\ \sigma = 0.04)$

Model parameters of the best-performing model (Zeppelin-and-Stick): $S_0 = 0.9830$, $d = 1.5835 \times 10^{-3}$, $\lambda_2 = 4.6759 \times 10^{-4}$, $f = 0.4354$

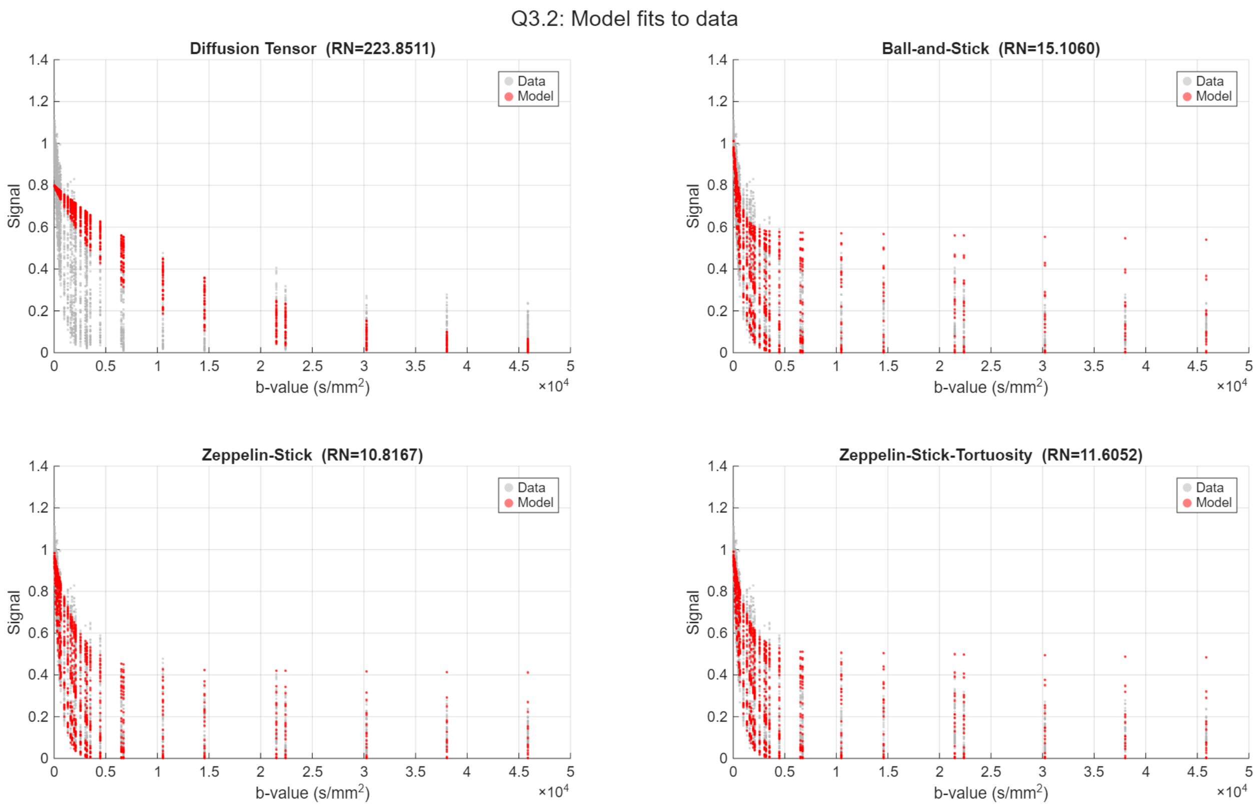


Figure 19: Model fits to the diffusion MRI data for the diffusion tensor, ball-and-stick, Zeppelin-and-stick, and Zeppelin-stick-tortuosity models. The residual norm (RN) is shown for each model.

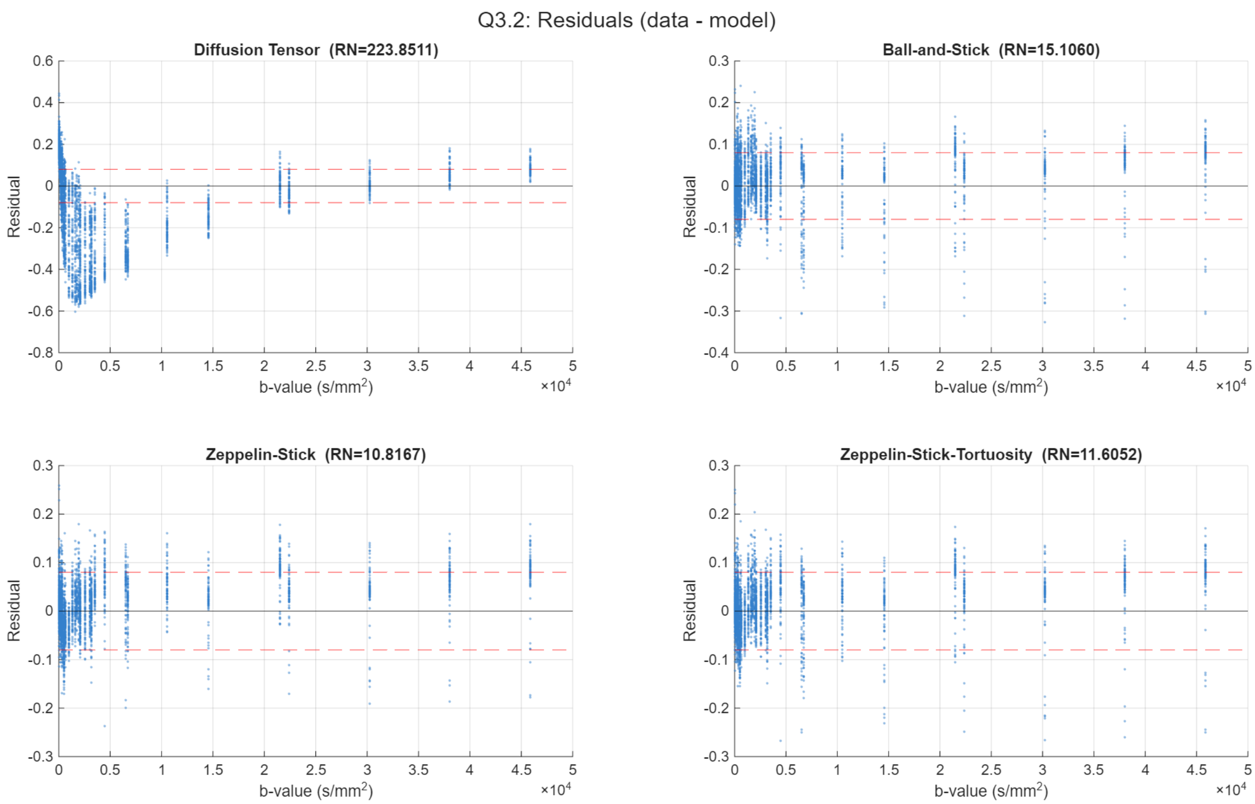


Figure 20: Residuals (data minus model) versus b-value for the diffusion tensor, ball-and-stick, Zeppelin-and-stick, and Zeppelin-stick-tortuosity models. The dashed lines show the $\pm 2\sigma$ noise bounds.

3.3 Ranking Models using AIC and BIC

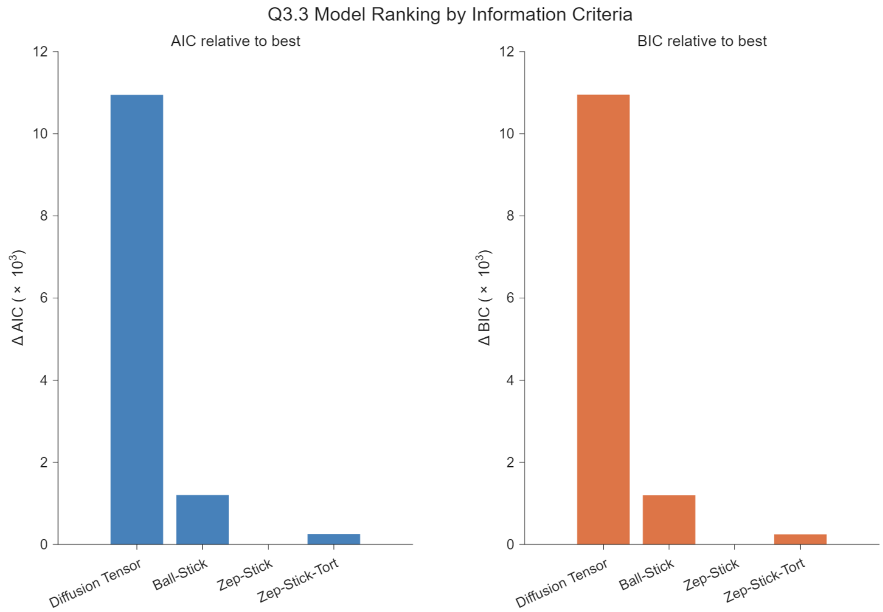


Figure 21: Model ranking based on information criteria. Bars show the difference in AIC and BIC relative to the best model. Lower values indicate better model support, with the Zeppelin–Stick model achieving the lowest score.

Table 13: Model selection results based on residual error and information criteria (Q3.3).

Model	N_{par}	RESNORM	SSD/K	AIC	BIC
Diffusion Tensor	7	223.8511	0.061974	-10029.10	-9979.56
Ball-Stick	5	15.1060	0.004182	-19770.65	-19733.50
Zep-Stick	6	10.8167	0.002995	**-20975.07**	**-20931.73**
Zep-Stick-Tort	5	11.6052	0.003213	-20722.91	-20685.76